

How loose is Wasserstein stability on a vortex field?

The Skraba–Turner cellular bound measured and certified on seven Gross–Pitaevskii pairs,
and where the duality points next

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Abstract

The previous report in this series [3] found that the bottleneck stability theorem for persistence diagrams, $d_B \leq \|f - g\|_\infty$, holds but is empty on six of seven pairs of two-dimensional Gross–Pitaevskii density fields: a few vortex-core pixels set the sup norm, and 2ε exceeds the entire diagram. It named, as its first direction, the stability theorem in which the sup norm is replaced by an ℓ_p norm, and set a gate: read the exact statement, fix every convention in advance, compute the right-hand side before looking at the left. This report passes that gate. The theorem is Skraba and Turner’s cellular Wasserstein stability [15] (Theorem 4.6): $W_p(\text{Dgm}f, \text{Dgm}g) \leq \|f - g\|_p$ with the norm taken over *all* cells of the complex, in the ℓ_p ground metric. We compute both sides exactly on the same seven pairs, for $p = 1, 2$: the right-hand side as a finite sum over the cubical complex whose cell values are verified, by an independent union-find, to be the ones GUDHI uses; the left-hand side as an optimal assignment on the full diagrams, certified on every pair by explicit dual potentials (rigorous relative gap $\leq 1.5 \times 10^{-4}$ on six certificates, $\leq 10^{-14}$ on the other twenty-two). Three controls pass exactly before any pair is opened, one of them an exact known answer on real data (slack exactly 9 for $p = 1$ and 3 for $p = 2$, from the nine cells a raised minimum touches), and one of them caught an arithmetic error in the author’s own pre-registered hand value. The pre-registered question is not whether the bound holds – it is a theorem – but whether it says anything the triangle inequality does not. The answer is no, on every pair: the ℓ_1 bound exceeds the trivial one by a factor V_1 of 139 (phase space) to 3200 (the vortex-free control), the ℓ_2 bound by 18 to 65, and the measured slack $S_1 = \|f - g\|_1/W_1$ runs from 490 to 19,000: in the theorem’s own metric the diagrams are three to four orders of magnitude closer than the fields. The pre-registered prediction that the phase-space pair – the one pair on which the bottleneck bound had been informative – would be informative here too, *failed*. The reason is structural and is measured, not argued: the ℓ_p norm is extensive in the number of cells, the diagram in the number of critical values, and a resolution or time comparison changes every cell; the ratio of the two bounds scales as $(\text{cells per bar})^{1/p}$, which is why $V_2 \approx \sqrt{V_1}$ on all seven pairs and why no p helps – small p pays the cell count, large p returns to the vortex cores. The theorem is sharp exactly where the control shows it: a perturbation that touches nine cells per critical value it moves. A second finding was not looked for: the previous report’s phase-space certificate had been computed on five points per side, the same class of defect as the corruption it reported; it is corrected here. Two further directions are taken to their gates and stopped there, with the reasons written down: the Berezinskii–Kosterlitz–Thouless transition, which these fields carry but which needs a finite-temperature run this project does not have; and the Fricke involution of an external review, which we now compare with Dolgachev’s theorem on the mirror moduli space of K3 surfaces read in full, and find to be the same involution without the group – a shadow, not an instance.

1 Introduction

A stability theorem for persistence diagrams is a promise: if two functions are close, their diagrams are close. The promise is only worth invoking if “close” on the right-hand side is a distance the data actually make small. On vortex fields the sup norm is not such a distance: a vortex core is a hole of depth n_0 a few pixels wide, and any comparison that moves a core by one pixel – a change of resolution, a step in time – pays the full depth in $\|f - g\|_\infty$. The previous report measured this on seven pairs: the bottleneck bound held on all seven with margins of 2 to 16, and excluded no bar at all on six of them [3].

The same report ended with three directions on *duality*, each behind a gate. The first was the obvious one: use the theorem whose right-hand side is an ℓ_p norm, since an ℓ_p norm averages over the field and a few core pixels no longer dominate it. The gate was to read the theorem before computing, and that gate did work: the draft attributed the ℓ_p bound to Cohen-Steiner, Edelsbrunner, Harer and Mileyko [5], whose bound in fact keeps the sup norm on the right (with a fractional exponent and a total-persistence constant), and the correction was made before publication. The theorem that changes the norm is Skraba and Turner’s [15], and it is the one tested here.

The question this report pre-registers (`docs/designs/WASSERSTEIN_STABILITY_PREREG.md`, committed before any code) is therefore not whether $W_p \leq \|f - g\|_p$ – it is a theorem, and a violation would be a bug in our pipeline – but two measurable things about it on real vortex fields:

1. *Is it informative?* The triangle inequality alone gives $W_p(\text{Dgm}f, \text{Dgm}g) \leq W_p(\text{Dgm}f, \emptyset) + W_p(\text{Dgm}g, \emptyset) =: Z_p$, the cost of erasing both diagrams. The theorem says something only where $\|f - g\|_p < Z_p$; we call $V_p = \|f - g\|_p / Z_p$ the *vacuity ratio*.
2. *How loose is it?* The *slack* $S_p = \|f - g\|_p / W_p$ measures how much closer the diagrams are than the fields, in the theorem’s own metric. It is a property of the data class, and it is what one would need to know to turn the theorem into a resolution criterion.

The second contribution is methodological and continues the previous report’s own lesson. There, a background agent corrupted a list of points between two pipeline stages, and only an independent recomputation caught it. Here nothing moves between agents: one script computes, its controls read the arrays from disk again, and the left-hand side is certified on every pair by a primal plan and dual potentials checked by a function that knows nothing about how they were found.

2 The theorem, with its conventions fixed

Theorem (Skraba–Turner, Theorem 4.6, Cellular Wasserstein Stability [15]). *Let $f, g : K \rightarrow \mathbb{R}$ be monotone functions on a finite CW complex K . Then*

- (i) $W_p(\text{Dgm}f, \text{Dgm}g) \leq \|f - g\|_p$, where $\|f - g\|_p^p = \sum_{\sigma \in K} |f(\sigma) - g(\sigma)|^p$;
- (ii) $W_p(\text{Dgm}_k f, \text{Dgm}_k g)^p \leq \sum_{\dim \sigma \in \{k, k+1\}} |f(\sigma) - g(\sigma)|^p$.

Three conventions travel with the statement and are adopted here as the theorem’s own metric, not ours. Distances in the plane are ℓ_p ; a point (a, b) is at perpendicular distance $2^{(1-p)/p}|b - a|$ from the diagonal ($|b - a|$ for $p = 1$, $|b - a|/\sqrt{2}$ for $p = 2$); essential classes are matched among themselves at cost $|a - b|$ and never to a finite point or the diagonal; and the total distance in (i) is $(\sum_k W_p(\text{Dgm}_k f, \text{Dgm}_k g)^p)^{1/p}$ over all degrees, each degree matched separately. This is *not* the ℓ^∞ ground cost of the previous report’s certificates; for $p = 1$ the ℓ_1 ground cost dominates it, so everything certified there is also bounded by the theorem, but the comparison is made in the theorem’s metric and the earlier numbers appear only as a consistency check (prediction P7).

The complex and the cell values. GUDHI’s periodic cubical complex on a 2D array with `top_dimensional_cells` is the T -construction [2, 16]: each pixel is a 2-cell with its own value and every lower cell takes the *minimum* over the pixels containing it. On an $N \times M$ torus that is

NM pixels, $2NM$ edges and NM vertices; along a non-periodic axis the boundary cells take the minimum over the pixels that exist. Both functions of every pair live on one such complex (the resolution pairs by the restriction rule of the previous report: the fine field restricted to the coarse grid), so the theorem applies literally, and $\|f - g\|_p$ is an exact finite sum over $4NM$ numbers.

3 Method and controls

Right-hand side. `cell_values` builds the four cell families from the padded array (wrap-around padding on a periodic axis, $+\infty$ padding otherwise) and `cell_norm` sums $|f(\sigma) - g(\sigma)|^p$ over all of them, or over vertices and edges only for part (ii) with $k = 0$.

Left-hand side. For each degree, the augmented $(n + m) \times (m + n)$ matrix of Kerber–Morozov–Nigmatov [9] with p -th powers of ℓ_p costs (each point owning one diagonal slot; cross-use forbidden by a large penalty; dummy-to-dummy free) is solved as an assignment (`scipy.optimize.linear_sum_assignment`); the sum is the p -th power of $W_p^{(k)}$. Then the dual LP $\max \sum u + \sum v$ s.t. $u_i + v_j \leq C_{ij}$ is solved (HiGHS, sparse; the dummy block replaced by two auxiliary variables and the penalty constraints dropped, which *relaxes* the dual and is therefore not trusted) and an independent function checks the potentials against the *full* matrix: feasibility everywhere and tightness against the plan. A plan with feasible, tight potentials is optimal by finite LP weak duality – the theorem proved in general in `lean_src/WassersteinCertificate.lean` of the previous report, which covers every certificate here. The diagrams are *full*: every finite bar and every essential class, not the top- k truncation used before.

Controls, all passed before any pair was opened.

- **C1** (the cell values are GUDHI’s). An independent union-find over the vertices and edges we build, with the elder rule, must reproduce GUDHI’s Dgm_0 exactly. Tested on random small arrays under all four periodicity patterns, with and without ties, and then on all twelve arrays of the seven pairs. All exact.
- **C2** (hand-computed known answer). The eight-point cycle of the previous report, $f = (1, 5, 2, 6, 3, 7, 4, 8)$ and g with $v_6 := 4.5$. Only one pixel changes and neither neighbouring vertex does (their minima are 3 and 4 on both sides), so $\|f - g\|_1 = \|f - g\|_2 = 5/2$; the diagrams differ in one point, $(4, 7) \rightarrow (4, 4.5)$, so $W_1 = 5/2$ and the $p = 1$ bound is *tight*. The script also had to return the previous report’s ℓ^∞ -ground $W_1 = 7/4$ on the same example when told to use that metric, as the check that the two conventions are the ones we think they are. It did.

Amendment A1, dated before any pair. The pre-registered hand value for W_2 on this example was $5/2$. The first control run returned $2.1506\dots$, and a second hand computation agrees: for $p = 2$ sending both points to the diagonal costs $((3/\sqrt{2})^2 + (1/2\sqrt{2})^2)^{1/2} = \sqrt{37/8}$, which is cheaper than the $5/2$ of matching them. The expected value was corrected to $\sqrt{37/8}$ and nothing else changed. The $p = 2$ bound holds on this example and is not tight. The control did what a control is for: it caught the author, not the code.

- **C3** (exact known answer on real data). Take ρ of pair T1’s first field, the birth pixel of its deepest finite H_0 bar that is a strict local minimum in 8-connectivity, and raise it by δ . All nine cells that pixel touches (itself, four edges, four vertices) have it as their minimum and change by exactly δ : $\|f - g\|_1 = 9\delta$, $\|f - g\|_2 = 3\delta$, the part-(ii) $k = 0$ sums are 8δ and $\sqrt{8}\delta$, and the diagram moves one birth by δ : $W_1 = W_2 = \delta$. Measured: slack exactly 9.000 and 3.000. The pre-registration allowed δ to be reduced if it exceeded the gap to the

pixel’s nearest neighbour; it did ($\delta = 0.01 \times \text{depth}$ exceeded the gap 4.3×10^{-3}) and δ was set to half the gap, as written, and recorded.

- **C4** (negative). Pair T1’s optimal plan with pair T2’s potentials must be rejected by the certificate checker. It was: maximal violation 0.898, primal 10.56 against dual 21.60, not feasible, not tight.

4 Results

Table 1 gives, for each pair and $p \in \{1, 2\}$: the theorem’s right-hand side $\|f - g\|_p$ over all cells; the part-(ii) bound for $k = 0$; the total W_p and its H_0 part $W_p^{(0)}$; the trivial bound Z_p ; the slack S_p and the vacuity ratio V_p .

pair	p	$\ f - g\ _p$	$B_p^{(ii,0)}$	W_p	$W_p^{(0)}$	Z_p	S_p	V_p	bound
R1	1	2.613e+05	1.964e+05	45.67	19.56	698.9	5.72e+03	374	holds
R1	2	364.9	316.8	1.47	0.979	20.13	248	18.1	holds
R2	1	3.712e+05	2.788e+05	55.46	23	868.7	6.69e+03	427	holds
R2	2	476.4	413.3	1.513	0.8816	21.95	315	21.7	holds
R3	1	4.847e+05	3.633e+05	79.58	28.79	1291	6.09e+03	376	holds
R3	2	593.1	513.3	1.716	0.8722	26.62	346	22.3	holds
T1	1	1.99e+06	1.495e+06	103	36.83	794.9	1.93e+04	2.5e+03	holds
T1	2	1205	1045	2.666	1.459	21.3	452	56.6	holds
T2	1	2.102e+06	1.578e+06	248	84.23	1109	8.48e+03	1.9e+03	holds
T2	2	1276	1106	4.722	2.343	24.82	270	51.4	holds
S1	1	1.695e+06	1.274e+06	285.5	167.5	530.4	5.94e+03	3.2e+03	holds
S1	2	1015	880.5	8.117	6.343	15.63	125	64.9	holds
P1	1	9707	7179	19.7	6.445	70.06	493	139	holds
P1	2	30.31	25.88	0.3871	0.2349	1.645	78.3	18.4	holds

Table 1: The Skraba–Turner bound on the seven pairs, in its own metric (ℓ_p ground cost, full diagrams, all degrees). $S_p = \|f - g\|_p / W_p$; $V_p = \|f - g\|_p / Z_p$, informative iff $V_p < 1$.

pair	p	rel. gap	max viol. δ	certified gap	scope	LP s
R1	1	0.0e+00	1.1e-16	8.5e-15	full	29.5
R1	2	1.8e-07	9.6e-08	1.5e-04	full	384.0
R2	1	0.0e+00	7.8e-17	6.2e-15	full	65.1
R2	2	3.3e-16	3.5e-18	7.9e-15	full	62.2
R3	1	1.3e-16	1.4e-17	9.0e-16	top-200	1.2
R3	2	1.1e-15	1.7e-18	1.1e-15	top-200	1.3
T1	1	1.9e-16	9.7e-17	4.4e-15	full	60.6
T1	2	2.2e-07	1.4e-07	1.2e-04	full	834.3
T2	1	4.9e-16	3.3e-16	6.7e-15	top-200	2.3
T2	2	0.0e+00	1.4e-17	2.2e-15	top-200	1.4
S1	1	1.7e-16	3.3e-16	2.7e-15	full	27.4
S1	2	0.0e+00	2.8e-17	8.8e-16	full	24.3
P1	1	2.6e-08	3.9e-08	4.2e-06	top-200	3.9
P1	2	4.5e-09	6.8e-10	5.0e-06	top-200	3.6

Table 2: Duality certificates for $W_p^{(0)}$: relative primal–dual gap as returned, maximal violation δ of $u_i + v_j \leq C_{ij}$ over the full augmented matrix, the rigorous certified relative gap after the $-R\delta$ shift, the scope (full diagram, or the 200 longest bars per side where the full LP exceeded the size rule of amendment A2), and LP time.

Pre-registered verdicts.

- **P1** (the bound holds, parts (i) and (ii)): holds on all seven pairs for $p = 1, 2$, by factors of S_p . As pre-registered, this line tests the pipeline, not the theorem.
- **P2** (resolution pairs vacuous at $p = 1$): **PASS** – $V_1 = 374, 427, 376$.
- **P3** (resolution pairs vacuous at $p = 2$ as well): **PASS** – $V_2 = 18.1, 21.7, 22.3$. With P2, the kill of the previous report’s §7.1 applies: no Wasserstein stability theorem makes the $\xi/\Delta x = 4$ vs. 8 comparison informative.
- **P4** (phase-space pair informative at $p = 1$): **FAIL** – $V_1 = 139, V_2 = 18.4$. The pair on which the bottleneck bound was informative ($2\varepsilon = 0.58$ against a persistence range of about 0.2 was the previous report’s finding; here $\varepsilon = 0.2878$ is reproduced) is not informative under the ℓ_p bound. See the mechanism below.
- **P5** (time pairs and the control, reported): $V_1 = 2500, 1900, 3200$ for T1, T2, S1; these are the largest, because the fields differ everywhere.
- **P6** (slack $S_1 \geq 10$ on every GP pair): **PASS**, by a wide margin – S_1 from 5720 (R1) to 19,300 (T1); S_2 from 125 to 452.
- **P7** (consistency with the previous report’s ℓ^∞ top-30 W_1): **PASS** on all seven; the six GP values are reproduced to all printed digits (2.3371, 0.5791, 0.8031, 2.4532, 0.9513, 10.8003), and ε likewise. The phase-space value is discussed in §5.2.

Why every pair is vacuous, measured. Write $\bar{\Delta}$ for the mean of $|f(\sigma) - g(\sigma)|$ over the $4NM$ cells and $\bar{\pi}$ for the mean persistence of the $n + m$ finite bars. Then $\|f - g\|_1 = 4NM \bar{\Delta}$ and $Z_1 = (n + m) \bar{\pi}$, so $V_1 = (4NM/(n + m)) \cdot (\bar{\Delta}/\bar{\pi})$: the number of cells per bar times the ratio of mean cell change to mean bar length. On R1, $4NM/(n + m) = 1.05 \times 10^6/1495 = 701$ and the measured $\bar{\Delta} = 0.249 n_0$ – the two resolutions are two different turbulent trajectories, not one field sampled twice, and they differ by a quarter of the mean density on the *average* cell. On T1 the mean cell change is $0.47 n_0$; on the vortex-free control S1, $0.40 n_0$. Nothing about vortex cores enters: the ℓ_1 bound is empty because the perturbation is dense, and a dense perturbation of a field with 10^3 cells per feature is charged 10^3 times per feature.

For general p the same count gives $V_p \sim (4NM/(n + m))^{1/p} \bar{\Delta}_p/\bar{\pi}_p$, and the data follow it: $V_2/\sqrt{V_1} = 0.94, 1.05, 1.15, 1.13, 1.18, 1.15, 1.56$ across the seven pairs. The limit $p \rightarrow \infty$ is the bottleneck case of the previous report, vacuous for the other reason (a few core pixels set the norm, and 2ε exceeds the diagram). There is no p at which a dense perturbation of a vortex field is small in ℓ_p and large in persistence: small p pays the cell count, large p pays the core depth.

Exploratory: is there a p that helps? Not pre-registered, computed after the verdicts above and labelled accordingly: the H_0 vacuity ratio $V_p = \|f - g\|_p/Z_p^{(0)}$ (all cells over the H_0 diagrams alone, primal only, no certificate) for $p \in \{1, 2, 4, 8\}$ and, at $p = \infty$, the bottleneck case $\varepsilon/(\max \text{persistence}/2)$ of the previous report.

pair	V_1	V_2	V_4	V_8	V_∞
R1	694	24.2	5.1	2.62	2.95
R2	869	31.1	6.12	2.94	3.52
R3	855	35.1	6.96	3.21	3.67
T1	4.92e+03	78.8	9.84	3.6	3.02
T2	4.19e+03	79.1	10.6	4.02	3.71
S1	6.1e+03	87.2	10.2	3.49	2.06
P1	387	30.5	5.95	2.4	3

V_p falls with p as the cell count is taken to the power $1/p$, reaches 2.4–4.0 at $p = 8$, and is 2.1–3.7 at $p = \infty$. Its minimum over p is above 2 on every pair. The two vacuity mechanisms – the cell count at small p , the core depth at large p – meet in the middle without ever crossing 1.

This is the measured form of the sentence in the abstract: on a dense perturbation of a vortex field, no ℓ_p stability bound says anything the triangle inequality does not.

What the slack measures. S_1 is between 5×10^3 and 2×10^4 on the GP pairs and 493 on the phase-space pair. The diagrams are not far apart: W_1/Z_1 , the actual distance as a fraction of the cost of erasing both diagrams, is 0.065 (R1), 0.064 (R2), 0.062 (R3), 0.13 (T1), 0.22 (T2), 0.54 (S1), 0.28 (P1). The resolution pairs' diagrams are within 7% of erasure of each other – close, by the standard the theorem uses – while their fields are not close at all. That is a statement about persistence as a summary, and it is the quantitative content of this round: on this data class the H_0 diagram is stable to dense perturbations to a degree the stability theorem cannot see, because the theorem is sharp only for sparse ones. The control C3 shows the sharp case exactly: slack 9 at $p = 1$ and 3 at $p = 2$, the coordination number of a pixel and its square root.

Part (ii) against part (i). Restricting to vertices and edges removes the pixel term and nothing else; on every pair $B_1^{(ii,0)}/B_1^{(i)}$ lies between 0.74 and 0.75 ($3/4$ would be exact if the three cell families changed equally), and $B_2^{(ii,0)}/B_2^{(i)}$ between 0.85 and 0.87. The per-degree refinement is real and small; it does not change any verdict.

5 Where the duality points next

5.1 What this bound can and cannot become

The kill written into the previous report is now executed: the resolution comparison $\xi/\Delta x = 4$ vs. 8 is not a perturbation in any ℓ_p sense, and no stability theorem of this family rescues it. The honest use of persistence at $\xi/\Delta x = 4$ is therefore not “the diagram is guaranteed close to the converged one” but “the diagram is measured to be within 7% of erasure of the converged one, with no guarantee”. Where the theorem *is* sharp – sparse perturbations, C3 – it is a tool for something else: certifying that a localized edit to a field (a planted vortex moved, a core filled in) changes the diagram by at most nine cells' worth, which is a statement one could use in a control experiment. That is the direction the bound can become, and it is small. The number to carry forward is the slack: on vortex fields, 10^3 – 10^4 at $p = 1$.

5.2 Erratum to the previous report

The previous report's Table 3 lists the phase-space pair's certificate as $n, m = 5, 5$ with $W_1 = 0.0632 \dots$. The full H_0 diagrams of the raw phase-space arrays have 2150 and 1623 finite bars, so the pre-registered top-30 truncation should have had 30 points per side. The diagrams stage of that round was correct – the bottleneck distance recomputed here on the full diagrams is $0.0325030 \dots$, the published $d_B = 0.0325$, and $\varepsilon = 0.2878$ is reproduced – and the 5×5 certificate is valid for the five points it was given; but the number is not the ℓ^∞ top-30 W_1 of the phase-space diagrams. It is the same class of defect as the S1 corruption that report found and fixed, on the one pair it did not independently recompute. The correct value, computed on the full diagrams in this round with the previous convention, is $W_1^{\ell^\infty, \text{top-30}} = 0.6490$. The published PDF is not changed; the erratum is recorded in the project ledger and here.

5.3 The transition these fields carry: BKT, and the run this project does not have

A vortex field in two dimensions is the setting of the Berezinskii–Kosterlitz–Thouless transition, where bound vortex–antivortex pairs unbind and the superfluid density jumps by the universal amount of Nelson and Kosterlitz [11], measured in helium films by Bishop and Reppy [1] and in

a trapped atomic gas by Hadzibabic et al. [8]. Persistence of the density field sees vortex cores as H_0 bars and, by the duality of constructions [2] used as a control in [4], nothing else; whether the *pairing* of cores – the quantity that changes at the transition – leaves a signature in Dgm_1 of the density or in the diagram of the phase field is a question with a known answer to test against: the critical density $n_c \lambda_T^2 = \ln(\xi/\tilde{g})$ with $\xi \approx 380$ of Prokof'ev, Ruebenacker and Svistunov [12].

Gate, and why it is not passed here. The fields of this and the previous report are zero-temperature Gross–Pitaevskii evolutions of planted vortices. BKT is a thermal transition; the standard numerical route is the projected Gross–Pitaevskii (classical-field) method, used for exactly this question by Simula and Blakie [14] and by Foster, Blakie and Davis [7], who resolve vortex-pair unbinding, the superfluid fraction and the first-order correlation function across the transition in a homogeneous box. This project has no finite-temperature solver. Writing one is a project of its own, with its own known answers (the superfluid jump, n_c), and it must be pre-registered as such; nothing in the present data can be reinterpreted as thermal. The direction stays open and unstarted.

5.4 The Fricke involution, read against Dolgachev

The external review quoted in [3] placed this project's older involution $R \leftrightarrow \alpha'/R$ – in `lean_src/DualLength.lean`, the map $k \mapsto k_s^2/k$ on wavenumbers, which exchanges the phonon and free-particle terms of the Bogoliubov dual length $\ell(k) = 1/k + k/k_s^2$ and fixes $k = k_s$ – with the Fricke involution and the Dolgachev–Nikulin mirror symmetry of K3 surfaces, as “a theorem” that “keeps the duality”. The gate set there was: read Dolgachev and decide whether the inversion is an instance of his theorem or its shadow. We read Dolgachev's paper [6] in full (arXiv:alg-geom/9502005). The relevant statements are:

Theorem 7.1. For the mirror family of degree- $2n$ polarized K3 surfaces (the M_n -polarized surfaces, $M_n = U \perp \langle -2n \rangle$ -orthogonal, Picard rank 19), the moduli space is $K_{M_n} \cong \mathbb{H}/\Gamma_0(n)^+$, where $\Gamma_0(n)^+$ is $\Gamma_0(n)$ extended by the Fricke involution $F = \begin{pmatrix} 0 & -1/\sqrt{n} \\ \sqrt{n} & 0 \end{pmatrix}$, $t \mapsto -1/(nt)$.

Theorem 7.6. An M_n -polarized K3 surface with period t is, up to a canonical involution, the Kummer surface of $E_t \times E_{-1/nt}$, a pair of n -isogenous elliptic curves; the Fricke involution acts on $\mathbb{H}/\Gamma_0(n)$ by $(E, A) \mapsto (E/A, E_n/A)$.

Now compare. Restrict F to the imaginary axis, $t = iy$: $F(iy) = i/(ny)$, i.e. $y \mapsto 1/(ny)$, an inversion of the positive reals about $y = 1/\sqrt{n}$. With $y = k$ and $n = 1/k_s^2$ this is exactly `DualLength`'s $k \mapsto k_s^2/k$ with fixed point k_s . So the reviewer is right that it is the same involution – and so is every inversion $x \mapsto c/x$, which is the point. What makes Dolgachev's statement a theorem about K3 surfaces is not F but the group $\Gamma_0(n)$ that F normalizes: the translation $t \mapsto t + 1$, the congruence condition $n \mid c$, the lattice-polarized family whose periods live in $\mathbb{H}$, and the isogeny that F realizes on the elliptic curves. A wavenumber k carries none of it: there is no $t \mapsto t + 1$ on the Bogoliubov dispersion, no lattice, no family whose monodromy is $\Gamma_0(n)$. `DualLength.lean` proves, and says it proves, an AM–GM statement about two positive terms.

Verdict at the gate: shadow, not instance. The honest sentence is “the same involution, without the group”. It is not a claim about helium, and no Lean line is written for it. What *would* be an instance is a system whose parameter space carries the modular action – the quantum Hall σ -plane under $\Gamma_0(2)$ [10], the Seiberg–Witten u -plane under $\Gamma(2)$ [13] – and those are the reviewer's other examples, outside this project's data. Formalizing Dolgachev's Theorem 7.1 itself in Lean 4 is out of reach of Mathlib as pinned here (no lattice-polarized K3 surfaces, no period map); formalizing the group-theoretic core – that F normalizes $\Gamma_0(n)$ in $\text{PSL}_2(\mathbb{R})$, that $\Gamma_0(n)^+$ is a group containing $\Gamma_0(n)$ with index 2 – is within reach and would be honest as *group theory*, labelled as such, with no claim of physics attached. It is recorded as a possible next Lean file and not started.

5.5 The certificate duality itself

The one duality that has done work in two reports is the humblest: finite LP weak duality, which turns a solver’s answer into a proof. Its next step is not new physics but a larger reach: the seven full-diagram certificates here are checked in Python by an independent function; the general theorem is in Lean; the specific 1600×1600 floating-point matrices are not re-verified by the kernel. Exact rational certificates for full diagrams, checked by `norm_num` or by a verified checker, would close that last gap and are the natural P6 of a future round.

6 Discussion

Three directions were named behind gates; this report is the record of taking each to its gate. The first passed and was measured: the cellular Wasserstein bound holds, as it must, and is empty on every pair, with the mechanism quantified (cells per bar) and a control showing where it is sharp. The pre-registered prediction that the phase-space pair would be informative was wrong, and is reported as wrong. The second and third stopped at their gates – one for want of a finite-temperature solver, one because the theorem it was compared with turned out to be about a group the data does not carry – and the reasons are written down, which is what a gate is for.

What this round did not do. One seed, one solver, one data class; $p = 1$ and 2 only, with the p -scaling inferred from two points and a count, not from a sweep; certificates on the full diagrams where the LP fit in memory and on the 200 longest bars where it did not, scope recorded per line of Table 2; no Lean line for Theorem 4.6 itself, whose statement is about persistence modules on a CW complex and is beyond the pinned library. And the round’s own small lesson: the author’s hand value for W_2 on an eight-point example was wrong, and the control caught it before any data was opened – the same discipline that found the previous report’s five-point certificate after publication rather than before. Redoing independently remains the only check that has worked.

Reproducibility. `exploration/tda/wp_stability.py` runs the controls and the seven pairs and writes `data/generated/kinetic_tda/wp_stability_results.json`; `wp_stability_report.py` produces the tables above from that file and re-derives the verdicts. Pre-registration and amendment: `docs/designs/WASSERSTEIN_STABILITY_PREREG.md`. The data arrays are the previous report’s (`make_gp_frames.py`, `make_gp_frames_n512.py`, seed as recorded there).

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